Supplementary Information for
When do single-cell response distributions improve drug retrieval?

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Supplementary Methods

Calculation of population retrieval scores. Energy distance was the primary population score. RBF-kernel maximum mean discrepancy used the median squared pairwise distance of the pooled query and candidate sample as its bandwidth reference and evaluated scales of 0.25, 1 and 4 times that value. Sliced Wasserstein-1 distance averaged quantile-matched projections over 128 fixed random directions. Candidate populations were subsampled to at most 500 cells in the observed-population analyses and 300 cells in the partial-observation analyses. A stricter task-specific cap takes precedence where reported in the main Methods. For a score with task seed s, the candidate and query populations were sampled with seeds s and $s + 1$, respectively. Analyses without replicate-specific task seeds used $s = 0$; partial observation used $s \in \{0, \ldots, 4\}$. A population containing fewer than two cells after filtering remained in the candidate library with distance 10^6. The same value is used for an empty or undersized candidate component in the coverage score.

Calculation of subpopulation coverage. In construction-labelled experiments, the known construction labels are used directly when they are carried by the task. In the controlled-mixture ranker, candidate labels are unavailable, so candidate cells are assigned to the nearest target-component mean by Euclidean distance. For partial observation, query cells are partitioned by k-means with $k = 2$, three initialisations and the task seed $s \in \{0, \ldots, 4\}$; candidate cells are assigned to the query component whose centroid has the highest cosine similarity after row normalisation. A candidate with no cell assigned to a query component received distance 10^6 for that component. At small β, the coverage distance was evaluated using $D_\beta = \text{mean}(d) + \frac{\beta}{2}\,\text{var}(d) + O(\beta^2)$ to avoid numerical cancellation in the log-sum-exp form. Coverage-mean and coverage-worst use the $\beta \to 0$ and $\beta \to \infty$ limits, respectively. The base distance was energy distance unless otherwise specified; the complete score registry is given in Supplementary Table 1.

Secondary retrieval summaries. In the observed-population task, query-weighted Hit@1 was 0.884 for energy distance and 0.421 for mean cosine, compared with the unweighted task–setting summaries reported in the main text. PCA-space population and mean retrieval reached 0.778 and 0.518, respectively. The both-mode candidate had no tied score at any rank across 1,260 queries and nine scorers, so optimistic, pessimistic and fractional tie handling gave the same Hit@1. WTCS gave Hit@1 of 0.4635.

For MoA recovery, excluding the 21 leave-drug-out queries without a same-mechanism candidate left the median population-minus-mean difference at 0.000 and changed the mean from -0.024 to -0.025. The decreased and increased shares changed from 40.8% and 35.7% to 42.3% and 37.0%, respectively. This comparison uses worst-case coverage against mean cosine.

Supplementary Note 1. Query-level predictors of population benefit

Pre-specified query-selection rule. The rule combined preference conflict with a structure-reliability score. Preference conflict was $(1 - \bar{\tau})/2$, where $\bar{\tau}$ is the mean Kendall correlation between candidate rankings computed separately for the two query states using energy distance. Structure reliability was $0.15\min(v/0.05, 1) + 0.10\min(d, 1) + 0.40b + 0.35\min(m, 1)$, where v is the fraction of query sum-of-squares explained by a two-cluster partition, d is one minus the mean pairwise cosine similarity among query cells, b is the fraction of five bootstrap resamples retaining the same top-ranked energy-distance candidate, and m is the Kendall distance between the mean-cosine and energy-distance rankings. All components lie in $[0, 1]$. Queries were selected when both scores were at least 0.4, assigned "mean sufficient" when structure reliability was at least 0.4 but preference conflict was lower, and otherwise assigned "mean or no call". The response-matching and mechanism-recovery outcomes were not used to construct the rule.

Response matching and mechanism recovery across scores. Across the five population-level scores, median response-matching gains over mean cosine ranged from $+0.028$ to $+0.067$, whereas mean mechanism-recovery gains ranged from -0.033 to -0.011 across the 600 leave-drug-out queries.

Alternative population scores in the observed-response task. Maximum mean discrepancy reproduced the energy-distance gain (MRR 0.975; $+0.0101$ over magnitude-aware mean L_2). Sliced Wasserstein reached 0.922, below directional mean cosine (-0.0230, 95% CI $[-0.0301, -0.0166]$) and magnitude-aware mean L_2 (-0.0431, $[-0.0509, -0.0359]$).

Energy-estimator sensitivity. The primary energy score excludes self-pairs through its U-statistic estimator. On the same observed-response queries, the V-statistic, which includes self-pairs, gave an MRR increment of $+0.0006$ over magnitude-aware mean L_2 (95% cell-line-cluster bootstrap CI -0.0019 to $+0.0028$), compared with $+0.0102$ ($+0.0065$ to $+0.0145$) for the U-statistic.

Headroom and conditional associations. Directional mean cosine ranked the true drug first at every seed for 3,110 of 3,992 queries. Energy retrieval lost 0.0036 on average in these queries and gained 0.1501 in the remaining 882. The headroom–gain correlation ($\rho = 0.79$) was reproduced by the ceiling-preserving permutation null (0.786, interval $[0.767, 0.805]$), so it did not identify population-beneficial queries (Fig. 5d,e).

After conditioning on directional mean-cosine reciprocal rank, response detectability was associated with energy retrieval ($+0.0091$, 95% CI $[+0.0060, +0.0121]$; 3,618 queries in 44 cell-line clusters). Neither the cross-fitted interaction ratio ($+0.0015$, $p = 0.21$) nor its reproducibility (-0.0011, $p = 0.21$) showed a supported association. Replacing the cross-fitted measure with naive response cosine produced a negative association (-0.0096, 95% CI $[-0.0167, -0.0024]$), demonstrating sensitivity to how interaction is measured (Fig. 5f–h).

Control-reference sensitivity. We repeated the rankings after subtracting matched control means, keeping queries, candidate libraries, seeds and evaluation fixed. Across 2,250 SciPlex3 cross-line queries, energy Hit@1 changed from 0.9189 to 0.9158 and MRR from 0.9478 to 0.9462. All population-minus-directional-mean effects remained positive; Frangieh IFNGR1 contrasts were unchanged. A common control translation leaves the controlled-mixture, GDSC2 and same-condition RNA population distances unchanged.

Under partial observation, control subtraction also changed query clustering and candidate assignment. Across 765 queries, mean regret reduction increased from 0.1320 to 0.2021 for worst-case coverage and from 0.1155 to 0.2041 for mean coverage. The top candidate changed in 88.5% and 86.7% of queries, respectively. Thus, response-matching gains persisted, but individual rankings were reference-sensitive.

Measured response divergence. Queries were stratified by response divergence, defined as one minus the mean cosine between state-specific responses. Mean mechanism-recovery gains across increasing quartiles were -0.0497, -0.0391, -0.0158 and $+0.0142$. The divergence–gain association was positive ($\rho = 0.1548$, 95% CI $[0.0722, 0.2349]$), but the highest-quartile gain remained uncertain (95% cluster-bootstrap CI $[-0.0165, +0.0425]$; per-query standard deviation 0.169). Neither this stratum nor the full 600-query comparison established a consistent mechanism-recovery advantage (Fig. 3b,d).

Query-selection comparison. Among the 720 individual-drug queries, the rule selects 584, assigns 125 to "mean or no call" and assigns 11 to "mean sufficient". The selected-minus-unselected median regret-gain difference is -0.0109 (95% drug-cluster bootstrap CI $[-0.0355, +0.0498]$; 584 versus 125 queries from 141 drugs), and pooling the 11 mean-sufficient queries with the unselected group leaves it unchanged. Selection therefore did not enrich for population-beneficial queries.

Supplementary Note 2. HIR-Bench construction

HIR-Bench compares candidate preferences under different population objectives. The analytical threshold below describes the mean-welfare comparison between two specialist candidates.

Each synthetic instance specifies two subpopulations: a majority with welfare weight α and a minority with weight $1 - \alpha$. In a 50-dimensional latent space, each subpopulation has an orthogonal target response. Candidate response centres form a continuum from minority-specialist to balanced to

majority-specialist, controlled by a conflict parameter in $[0, 1]$; remaining candidates are distant distractors. Candidate cells are Gaussian draws around their subpopulation-specific centres. The oracle utility of candidate d for subpopulation k is $u(d, k) = -\mathbb{E}\|X_{d,k} - t_k\|_2^2$, where t_k is that subpopulation's target. Let d_M be the candidate with maximal utility in the majority and d_C the candidate with maximal utility in the minority. When $d_M \neq d_C$, define

$$A = u(d_M, 0) - u(d_C, 0), \qquad B = u(d_C, 1) - u(d_M, 1). \tag{S1}$$

The majority-weighted utility difference is $\alpha A - (1 - \alpha)B$, so its sign changes at

$$\alpha^* = \frac{B}{A + B}. \tag{S2}$$

Thus α^* is the analytical boundary at which the majority-weighted preference switches between the two conflicting candidates; it is undefined and not evaluated when there is no preference conflict.

The benchmark grid uses $p = 50$, $\alpha \in \{0.5, 0.7, 0.85, 0.95\}$, conflict $\gamma \in \{0, 1/6, \ldots, 1\}$, $n \in \{15, 60\}$ cells per candidate subpopulation, $\sigma \in \{0.3, 0.8\}$, library size $L \in \{20, 100\}$ and seeds 0–19. Two target vectors are drawn from standard normals, orthogonalised by QR and scaled to norm $\sqrt{p}$. The $L - \max(1, \lfloor L/5 \rfloor)$ structured candidates have evenly spaced biases $b_d \in [-1, 1]$. With $e_d = b_d\gamma$, a random unit direction v_d shared by their two states and $a = 0.3\sqrt{p}$, their centres are $\mu_{d,0} = t_0 + a(1 - 0.5e_d)v_d$ and $\mu_{d,1} = t_1 + a(1 + 0.5e_d)v_d$. Distractor centres are independent draws from $\mathcal{N}(0, 4pI)$. Candidate cells follow $\mathcal{N}(\mu_{d,k}, \sigma^2 I)$, with exactly n cells in each state; oracle utility is $-\|\mu_{d,k} - t_k\|^2 - p\sigma^2$.

Query state k contains $\max(5, \lfloor 2nw_k \rfloor)$ draws from $\mathcal{N}(t_k, (0.8\sigma)^2 I)$, with $w_0 = \alpha$ and $w_1 = 1 - \alpha$. Thus the welfare weights are specified independently of the equal candidate-state cell counts. The three information conditions retain observed candidate cells, replace them by draws around their empirical pooled mean with variance $\sigma^2 I$, or reconstruct them by PCA with $\min(\max(2, \lfloor p/5 \rfloor), \min(N, p) - 1)$ components and add Gaussian noise of standard deviation 0.3σ; PCA uses seed 0. The full grid contains 13,440 instances, evaluated under mean and worst-state welfare.

Supplementary Note 3. Robustness of the natural-heterogeneity and large-scale analyses

Response alignment between compartments. For 17 patient–drug pairs in which both compartments carried enough treated and control cells, the cosine between the malignant and myeloid perturbation-response directions has a median of 0.566, against a range of 0.02 to 0.39 for the constructed mixtures used elsewhere in this study. Response-direction cosine varies with effect size, signal-to-noise and cell number.

In the one patient with six jointly eligible drugs, each drug supplied a query with five alternatives. All six queries selected different closest response matches in the malignant and myeloid compartments, despite a median per-query rank correlation of 0.850. Each of the other three patients had two eligible drugs, giving a single alternative per query.

Robustness of glioblastoma compartment assignment. Malignant, tumour-associated myeloid and oligodendrocyte compartments were assigned from marker scores. Cells with fewer than 200 detected genes were removed, counts were library-size normalised to 10^4 and $\log(1 + x)$ transformed. Compartment scores were calculated on the full matrix before the 2,000-gene selection. Three marker sets were used: malignant *SOX2*, *EGFR*, *PDGFRA*, *OLIG1*, *OLIG2*, *GFAP* and *AQP4*; myeloid *PTPRC*, *CD14*, *AIF1*, *CSF1R*, *ITGAM*, *C1QA* and *C1QB*; oligodendrocyte *MBP*, *PLP1*, *MOG* and *MAG*. Each gene was z-scored across all cells of the cohort and the compartment score is the mean of its set's z-scores. A cell was called for the highest-scoring compartment only if the mean log-normalised expression of that compartment's key markers (malignant *SOX2*, *OLIG2*, *EGFR*; myeloid *CD14*, *CSF1R*, *C1QB*, *AIF1*; oligodendrocyte *MBP*, *PLP1*) reached 0.25 and its z-score margin over the second-highest compartment reached 0.25. Cells failing either criterion were left unassigned, leaving 39.9% of cells unassigned. Every called compartment was required to express its own key markers most strongly.

The results were stable when both thresholds were varied jointly over

$$\{0.10,\ 0.25,\ 0.40,\ 0.50\}. \tag{S3}$$

Across this range, the excluded fraction varied from 29% to 58%, while the supervised drug-response ceiling remained between 0.915 and 0.923, the best unsupervised recovery between 0.77 and 0.80, and the paired recoverability gap between approximately 0.11 and 0.13. The median cosine between malignant- and myeloid-specific perturbation-response directions remained between 0.54 and 0.59 across thresholds.

Drug-response recoverability. Within each patient and compartment, two treated populations were balanced by sampling up to 150 cells per drug, requiring at least 50 per drug, across five seeds. Both arms used the same up-to-30-component PCA fitted to the pooled, unlabelled mixture. Supervised accuracy was estimated by five-fold stratified cross-validation of logistic regression, random-forest and nearest-neighbour probes; scaling was fitted within each training fold. Unsupervised methods were k-means with $k = 2$, a two-component Gaussian mixture, Leiden[1] and HDBSCAN[2], evaluated by best-permutation accuracy. Leiden gave the highest unsupervised accuracy on 152 of the 180 natural splits. The primary comparison uses the best tested probe and clusterer; fixed-method comparisons assess sensitivity to this selection. Supervised accuracy provides an empirical reference for recoverability.

Recoverability of constructed response states. Supervised accuracy was 0.692 against 0.674 for the best unsupervised method in the class-pooled mixture, and 0.879 for individual drug-pair comparisons. Across nine alternative clustering configurations, median adjusted Rand indices ranged from 0.037 to 0.106. These comprised k-means with $k = 2$ in raw, 10-component PCA, 50-component PCA and response spaces; two-component Gaussian mixtures in raw and 50-component PCA spaces; and silhouette-selected $k \in \{3, 4, 5\}$, reported separately by the selected k.

In a sensitivity analysis over 20 seeds, increasing cell number from 25 to 400 per source changed the k-means adjusted Rand index by at most 0.02 at fixed separation. Increasing source separation from one to eight times its observed value raised the index from 0.07 to 1.00. Separation was rescaled using the full source populations before subsampling. More cells did not compensate for closely overlapping response states.

Patient-level uncertainty. The natural-tissue recoverability analysis contains 36 within-patient drug-pair splits across four patients and two compartments, with 30 of the 36 splits contributed by a single patient. Uncertainty was estimated by resampling patient clusters, and the analysis was repeated after removing the dominant patient.

The reported recoverability gap is computed within each split before aggregation,

$$G_j = A^{(j)}_{\text{sup}} - A^{(j)}_{\text{unsup}}, \tag{S4}$$

with a primary paired median of $+0.117$. Across 5,000 patient-cluster bootstrap resamples, the natural-minus-constructed difference in recoverability gap was $+0.109$ with an interval of $[+0.052, +0.129]$.

For each fixed probe–clusterer pair, seed repetitions are first collapsed within each split, and the paired gap is then summarised by its median across splits. The median across the nine combinations of three supervised probes with k-means, Gaussian-mixture or Leiden clustering is $+0.122$ in tissue against -0.002 in the constructed arm. Adding the three HDBSCAN pairings gives medians of $+0.193$ and $+0.020$ across all twelve combinations, respectively; HDBSCAN's median accuracy was 0.500 in both arms. The best-unsupervised comparison includes all four clusterers. Removing the patient that contributes 30 of the 36 splits gives $+0.123$, with a $[+0.098, +0.138]$ interval.

Large-scale measurement in Tahoe-100M. Tahoe-100M plate 3 measures intrinsic heterogeneity within cell lines. Two cellular-state definitions were specified independently of retrieval outcomes. The first is the released cell-cycle phase call, G1 against G2M, with S-phase cells dropped as intermediate and each

state's response referred to its own phase-matched control. The second is k-means with $k = 2$ fitted on each cell line's DMSO_TF control cells in the 2,304 highly variable genes in log1p space, with ten initialisations and seed 0; treated cells are assigned to the nearest control centroid, the larger control cluster is taken as the first state, and a cell line enters only when it holds at least 100 control cells. Across 3,630 evaluable drug–cell-line conditions, the median cosine between the two state-specific response directions was 0.739 (interquartile range 0.600–0.836). Fewer than 0.11% of conditions were as divergent as the constructed mixtures used in the controlled experiments.

Conditions were excluded when a required state-specific treated or control arm contained fewer than 50 cells, a filter that can preferentially remove strong cell-cycle-arresting perturbations. Across drugs, coverage was weakly associated with induced-response cosine (Spearman $\rho = 0.077$), and restricting the analysis to well-covered compounds gave a median of 0.740.

Recoverability remained measurable at scale. Across 960 sampled drug pairs and 48 cell lines, the supervised ceiling was 0.853 and the best unsupervised procedure reached 0.611, giving a median paired gap of 0.157. The gap was positive in all 48 evaluable cell lines. Individual unsupervised methods gave similar results: k-means 0.592, Gaussian mixtures 0.582 and Leiden[1] 0.563.

State-specific drug-response ordering. We finally compared relative drug-response ordering between non-overlapping cellular states. Under the cell-cycle partition, the median disjoint Spearman correlation between the two drug–drug similarity structures was 0.841 across 44 evaluable contexts. Under the independently control-derived two-state partition, the corresponding median was 0.778 across 45 contexts.

These correlations summarise the global drug-pair similarity structure across states. Under the control-derived partition, 2 of 45 contexts had a correlation below 0.5, with a minimum of 0.236. Direct query-level ranking measures which response analogues are selected within each context.

The two state definitions gave a Spearman correlation of $\rho = 0.466$ between their context rankings.

Using every jointly eligible drug as a query against the remaining drugs gave 3,620 cell-cycle queries in 44 cell lines and 3,722 control-derived-state queries in 45 cell lines. Median within-cell-line first-ranked agreement was 42.4% and 39.3%, respectively; the corresponding median top-five overlap was 58.5% and 51.7%. The median of within-cell-line median per-query rank correlations was 0.866 and 0.810, respectively. Thus, high agreement in the global similarity structure coexisted with frequent changes in the closest response match. These state comparisons use their own joint-coverage filters; intervention identification in Figs. 5 and 6 uses 3,992 queries and a fixed 92-compound library.

Supplementary Note 4. Perturbation-predictor diagnostics

The predictor analyses ask whether a predicted response population contains state-specific perturbation effects that a downstream population-aware retriever could exploit.

Additive response models. For a predictor that applies the same perturbation effect δ_d to every cell in a context,

$$\widehat{x}_{i,d} = x_i + \delta_d, \tag{S5}$$

the response of any two cellular states is identical after subtraction of their corresponding controls,

$$r_{d,0} = r_{d,1} = \delta_d. \tag{S6}$$

For a non-zero perturbation effect, consequently,

$$\cos(r_{d,0}, r_{d,1}) = 1. \tag{S7}$$

This result is independent of the heterogeneity of the starting control population. An additive predictor can therefore generate a visually heterogeneous predicted population while containing no state-specific variation in the perturbation effect itself.

The average-effect and linear-latent predictors used in the main analysis satisfy this additive limit by construction when the perturbation response is referenced to the model-consistent baseline.

Prediction-based retrieval in SciPlex3. In the SciPlex3 cross-context retrieval diagnostic, responses to each of ten divergent drugs were mixed across three cell-line pairs at three mixing fractions and eight seeds, giving 720 queries. Candidate responses were predicted from the corresponding mixed control populations and ranked by mean cosine, mean-delta R^2, energy distance or coverage. Evaluation used nDCG@10 with drug identity as binary relevance. Average-effect, nearest-neighbour and linear-latent predictors used all SciPlex3 contexts and therefore provided in-sample references; entropic optimal transport excluded the query context. Sampling, library construction, model parameters and seeds are specified in Supplementary Table 4.

Separating perturbation effects from reconstruction error. Subtracting a model's own zero-effect reconstruction isolates the induced response from reconstruction error. For an additive predictor $\hat{x}_{i,d} = g(x_i) + \delta_d$, this gives identical non-zero response vectors across states and cosine 1. Subtracting real controls instead also includes state-dependent reconstruction error.

Across 30 cell-line–drug combinations, induced-response cosines against the model baseline were 1.000, 1.000, 0.972 and 0.900 for average-effect, linear-latent, scGen and optimal-transport predictors, respectively. Against real controls they were 1.000, 0.190, 0.404 and 0.380; observed treated cells gave 0.205. The low real-control cosine of the additive linear-latent model therefore reflected reconstruction error, not a recovered state-specific perturbation effect.

scGen. scGen[3] was fitted using scvi-tools[4]. A drug-specific latent shift was added to each encoded control cell before decoding (Supplementary Table 4). Its induced-response cosine of 0.972 against its own reconstruction was close to the additive limit, indicating less state dependence than observed in treated cells.

Entropic optimal-transport mapping. We additionally evaluated an entropic optimal-transport map as a predictor with cell-dependent displacements. It is a different map from the Gaussian (Bures) predictor of the Tahoe-100M arm (Methods). Under the within-context differential-response analysis, the optimal-transport predictions reached an induced-response cosine of approximately 0.900 ± 0.107 against their own reconstruction baseline, moving away from the additive limit but remaining much more aligned than the observed responses.

The entropic map was fitted using Sinkhorn optimisation[5] in POT[6]. It used a 30-component PCA representation, with the query context excluded from fitting; full parameters are given in Supplementary Table 4.

In the cross-context predict-then-rank setting, the same entropic map gave Hit@1 values of 0.053 under mean cosine and 0.007 and 0.008 under the two distributional rules, compared with 0.879 to 0.899 for the additive average-effect candidate source. The predictions therefore did not preserve retrieval performance in that transfer setting.

In the Tahoe-100M task, energy retrieval showed no supported MRR gain over directional mean cosine for any of the six predictors. Against magnitude-aware mean L_2, Gaussian optimal transport gave a gain of $+0.0110$ (95% cell-line-cluster bootstrap CI $+0.0087$ to $+0.0134$). Supplementary Table 6 reports both mean references for all six predictors.

Supplementary Table 1. Retrieval scores, task constructions and evaluation criteria.

(a) Retrieval scores.

Score	Input representation	Ranking rule	Role
Mean cosine	Control-referenced mean perturbation signatures	Descending cosine similarity between query and candidate mean-response vectors	Direction-only mean reference
Magnitude-aware mean L_2	Control-referenced mean perturbation signatures	Ascending Euclidean distance between query and candidate mean-response vectors	Mean reference retaining both response direction and magnitude
CMap-style cosine	Same control-referenced mean differential-expression vectors as mean cosine	Same cosine operation as the implemented mean-cosine baseline	Signature-retrieval reference; numerically identical to mean cosine under the specified scoring definition
WTCS	Mean-collapsed differential-expression response	Rank-based weighted connectivity / enrichment score	Canonical rank-based signature baseline
Energy distance	Raw normalised-expression single-cell response populations	Candidates ranked by increasing U-statistic energy distance to the query population	Primary population-level retrieval score
RBF-MMD	Raw normalised-expression response populations	Candidates ranked by increasing U-statistic maximum mean discrepancy under an RBF kernel	Alternative distribution-sensitive score
Sliced Wasserstein	Empirical response populations	Candidates ranked by increasing sliced Wasserstein distance using 128 random projections	Alternative distribution-sensitive score
Subpopulation coverage	Response populations decomposed into response components	Component-level distances aggregated with temperature β; increasing β gives progressively greater weight to the least well-matched response component	Population-sensitive score used for the partial-observation response-matching analyses

The four score families become five population-level configurations because subpopulation-coverage contributes two: coverage-mean is the $\beta \to 0$ limit of the aggregate and coverage-worst the $\beta \to \infty$ limit, both with energy distance as the per-component base metric and the same U-statistic default for energy/MMD components. The PCA-latent population score uses the V-statistic form.

Supplementary Table 1 continued. (b) Retrieval task constructions.

Task	Query and candidate construction	Candidate evaluation	Statistical unit
Observed-population retrieval over heterogeneous queries	A heterogeneous query population combines two response modes at weight α; the library holds a held-out re-mixture at the same weight, the two single-mode populations, and 40 distractors, giving a 43-candidate library	Hit@1; only the both-mode candidate is relevant, so a single-mode population counts as a miss even when it carries one of the query's two response modes	Retrieval query; performance macro-averaged over seven task–setting combinations
Controlled heterogeneous queries	Two response modes are combined at mixing weight α; controlled SciPlex3 analyses use HDAC- and JAK-associated response sources	Source identification and response-matching criteria	Query / task setting
Partial-observation retrieval	Candidate libraries follow the partial-observation protocol; all scores, including direction-only mean cosine and magnitude-aware mean L_2, use the same response information and candidate library	Each ranking is evaluated by response matching and MoA recovery; the mean-based minority-state similarity score is compared separately with a cell-level measure	765 total queries; formal inference clusters 720 individual-drug queries by 143 query drugs; 45 pooled partial-library queries are descriptive only
Leave-drug-out MoA analysis set	Same partial-observation rankings; all 600 leave-drug-out queries	MoA-nDCG@10	600 leave-drug-out queries from 130 query-drug clusters, including 21 zero-tie queries without a same-mechanism candidate; 120 leave-MoA-out and 45 partial-library queries are excluded because MoA recovery is undefined
Controlled-mixture validation of the state-similarity measure	K562 HDAC–JAK controlled mixtures at $\alpha = 0.7$, with 43 candidates under each of 20 fixed seeds	Mean-based minority-state similarity proxy versus a cell-level nearest-neighbour reference	860 candidate cases; top-ranked candidate compared within each seed
GDSC2-linked retrieval	Held-out SciPlex3 compound at 10 μM used as query; remaining matched compounds form a 30–34 evaluable-candidate library after valid-pair filtering	Spearman agreement between RNA-derived ranking and external GDSC2 functional-similarity ordering	103 query correlations pooled descriptively; cell-line-specific summaries retain the three higher-level units
Tahoe intervention identification (Figs. 5 and 6)	Vehicle cells and an observed treated population define each query; 92 candidates have either observed target-context responses or predictions fitted without that context	MRR and top-ranked intervention identity	3,992 queries in 44 cell-line clusters, each evaluated at five fixed seeds

Supplementary Table 1 continued. (c) Evaluation criteria applied to fixed rankings.

Criterion	Definition	Interpretation
Observed-population Hit@1 (Fig. 2)	Indicator that the both-mode candidate, the held-out re-mixture at the query's own weight, is ranked first	Recovery of the query's two response modes at their mixing weight
Intervention-identification MRR (Figs. 5 and 6)	Mean reciprocal rank of the query drug in the candidate library	Recovery of intervention identity from observed or predicted candidate responses
Response-matching regret	Energy-based welfare regret relative to the best available candidate, with utilities calculated separately for the response states of the controlled partial-observation task; reported relative to a named mean reference: direction-only mean cosine in Figs. 2c and 3a, and direction-only mean cosine or magnitude-aware mean L_2 separately in Fig. 2e	Response fidelity; intentionally close to the population information used by the retriever
MoA-nDCG@10	Normalised discounted cumulative gain through rank 10, with binary relevance for candidates sharing the query perturbation's annotated mechanism, logarithmic rank discount and ideal-ranking normalisation (Methods)	Task-proximal biological criterion
Mean-based minority-state similarity proxy	Cosine between candidate-mean and minority-state-mean responses, rescaled to $[0, 1]$	Mean-response diagnostic compared with a cell-level preservation measure (Fig. 3c)
GDSC2 functional similarity	Spearman correlation between cell-line-centred cross-cell-line AUC profiles of the query and candidate drugs	Similarity of drug-response profiles across cell lines
Protein direction-only evaluator	Cosine similarity between control-referenced mean protein-response vectors	Direction-only reference evaluator
Protein magnitude-aware mean evaluator	Negative Euclidean distance between control-referenced mean protein-response vectors	Primary mean-form evaluator matched to the Euclidean geometry of the population comparison
Protein population-energy evaluator	Negative energy distance between control-referenced per-cell protein-response populations	Primary population-form evaluator retaining cell-to-cell protein-response structure
Response-magnitude matching	$-\left\lvert \lVert \bar{d}_q \rVert - \lVert \bar{d}_c \rVert \right\rvert$	Query-dependent scalar control that uses no within-population information

Query states s are obtained by k-means with $k = 2$ on the query cells, the minority state being the smaller cluster. A candidate d with response population P_d receives the welfare $U(d) = \mathrm{agg}_s\left(-\mathcal{E}(P_d, Q_s)\right)$, where $\mathcal{E}$ is the energy distance and the aggregator is the minimum over states in the worst-case form and the mean in the mean form. The response-matching regret of a selected candidate $\hat{d}$ is $\mathrm{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0$, taken over the observed library $\mathcal{L}$, so the reference is the best available candidate in the library. The mean-based proxy is $\frac{1}{2}\left(1 + \cos(\bar{P}_d, \bar{Q}_{\min})\right)$, the cosine between the candidate's mean response and the mean response of the query's minority state rescaled to $[0, 1]$. The same definitions and the cell-level validation measure are given in the main Methods.

Supplementary Table 2. Data resources, preprocessing and analytical roles.

Resource	Scale	Perturbation/context structure	Representation and controls	Role in this study
SciPlex3	276,325 cells; 2,000 genes	188 compounds shared across A549, K562 and MCF7	Log-transformed 2,000-gene representation; mean signatures are referenced to matched controls, whereas population scores use normalised treated-cell populations	Primary observed-population drug-retrieval setting; controlled within- and cross-context constructions; source of the GDSC2-linked transcriptomic rankings
Frangieh Perturb-CITE-seq	218,023 RNA-profiled cells for retrieval and 218,331 barcode-matched cells for the RNA–protein evaluation; 2,000 genes; 20 retained surface proteins	239 CRISPR knockouts across Control, IFNγ and Co-culture conditions	RNA and protein modalities joined by cell barcode; mean signatures are referenced to condition-matched controls, whereas population scores use normalised treated-cell populations	Independent genetic-perturbation retrieval setting and matched RNA–protein evaluator-shape experiment
ZhaoSims2021 glioblastoma	10 specimen identifiers; 6 drug treatments	Acute-slice culture and surgical biopsy with co-existing malignant, myeloid and oligodendrocyte compartments	Counts normalised to 10^4 per cell, log1p-transformed and reduced to 2,000 highly variable genes; state-specific responses referenced within patient and compartment	Natural-tissue analysis of differential response, recoverability and state-specific response ordering
Tahoe-100M plate 3	4,158,278 cells; 2,304 genes	50 cell lines; 92 compounds at one concentration plus DMSO_TF vehicle controls	Released 2,304-gene log1p representation used without refitting; state-specific responses referenced to matched state controls	Two roles: large-scale measurement of differential response, recoverability and state-ordering agreement under intrinsic within-cell-line heterogeneity, and the substrate for the source-to-target intervention retrieval benchmark (Figs. 5 and 6)
GDSC2	286 compounds across 969 cell lines in the retained release	35 compounds matched to the SciPlex3 panel; A549, K562 and MCF7 excluded from external profiles	Fitted dose–response AUC; primary functional similarity uses cell-line-centred cross-line response profiles	External functional evaluation of fixed SciPlex3 RNA-derived rankings
HIR-Bench	Synthetic construction	Two subpopulations with majority weight α and a conflict parameter between specialist candidates	50-dimensional latent space; Gaussian candidate cells around subpopulation-specific centres	Mixture-weighted mean-welfare preference switch between two specialists (Fig. 4d)

Dataset-specific cell-number thresholds, subsampling rules and statistical units are given in the corresponding Methods sections.

Supplementary Table 3. Sensitivity of the GDSC2 functional-similarity evaluation to response-profile construction.

The primary external functional criterion compares drug–drug response profiles after removing the shared cell-line sensitivity effect. For cell line l and drug d,

$$\widetilde{\mathrm{AUC}}_{d,l} = \mathrm{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \mathrm{AUC}_{d',l}, \tag{S8}$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in that cell line. Drug–drug functional similarity is then the Spearman correlation between centred cross-cell-line response profiles. The table compares this primary construction with the corresponding uncentred AUC-profile similarity.

RNA-derived ranking rule	Cell-line-centred functional similarity	Uncentred functional similarity
Energy distance	+0.265	+0.0902
Magnitude-aware mean L2	+0.285	+0.0922
Mean cosine, control referenced	+0.083	+0.138
Mean cosine, treated mean only	+0.241	+0.092
Response-magnitude matching	+0.232	+0.217
Candidate magnitude only	−0.330	+0.135
Potency matching diagnostic	+0.399	+0.290
Energy, magnitude-adjusted partial rank correlation	+0.105	n/a

Values are median per-query Spearman correlations over 103 leave-one-drug-out queries across the three SciPlex3 cell lines. The primary centred functional criterion is constructed from GDSC2 AUC profiles after subtracting each cell line's mean response across all 286 compounds in the retained release. Drug pairs sharing fewer than 100 cell lines are excluded; 591 usable drug pairs remain, with a median of 909 shared cell lines per pair.

The median within-query difference (energy minus mean L2) is $+0.0012$ overall, -0.0032 in A549, $+0.0079$ in K562 and $+0.0024$ in MCF7. Energy is higher on 55 queries, mean L2 on 47, and one query ties. These paired medians differ from differences between marginal medians. Under the uncentred criterion, control-referenced mean cosine exceeds energy retrieval.

Supplementary Table 4. Perturbation-predictor implementations and their role in the differential-response analysis.

Predictor	Model form	Predicted-population construction	Use in this study	Response properties
Average-effect	Per-drug mean perturbation delta	Predicted delta applied to the real control cells of the query context	Predict-then-rank candidate source and within-context differential-response reference	Additive by construction; state-specific induced-response cosine equals 1
Nearest-neighbour	Transfers the perturbation effect from the training context with the most similar untreated control state	Transferred effect applied to real query-context control cells	Predict-then-rank candidate source	One transferred effect applied to every source cell
Linear-latent	Additive latent arithmetic in an encoded representation followed by linear decoding	Observed control cells encoded, shifted in latent space and decoded individually	Predict-then-rank candidate source and additive latent reference	Baseline cellular heterogeneity with an additive perturbation effect
scGen	Variational-autoencoder perturbation model with latent arithmetic	Evaluated on the same within-context state partition and state-specific response definition used for other predictors	Additional differential-response diagnostic reported in Supplementary Note 4	Mean induced-response cosine 0.972 over 30 cell-line and drug combinations
Entropic optimal-transport map	Non-parametric Sinkhorn mapping that permits cell-dependent displacements	Control cells mapped individually to predicted treated states	Additional non-additive differential-response diagnostic; also tested as a candidate-response source	Cell-dependent displacements; within-context induced-response cosine approximately 0.900 ± 0.107 against the map's own reconstruction baseline and 0.380 against real controls

Supplementary Table 4 continued. Implementation of each predictor.

Predictor	Implementation
Average-effect	Per drug d, $\widehat{\delta}_d = \text{mean}_c\left(\text{mean}(\text{treated}_{c,d}) - \text{mean}(\text{control}_c)\right)$ over the SciPlex3 cell lines c, a line contributing only where it has at least two treated cells for d; one signature per drug, independent of the query context. The predicted population is that single signature added to each of the query context's real control cells, so every cell is shifted identically. The zero-effect baseline is the real control population. A drug absent from the fit is scored at -10^6 under every rule
Nearest-neighbour	$c^* = \arg\min_c\left(1 - \cos(\bar{x}^{\text{query}}_{\text{control}}, \bar{x}^{c}_{\text{control}})\right)$ over the training lines, then $\widehat{\delta}_d = \text{mean}(\text{treated}_{c^*,d}) - \text{mean}(\text{control}_{c^*})$. Where d was not measured in c^* the signature falls back to the mean of d's signatures over the remaining admissible lines. Population synthesis as for average-effect
Linear-latent	PCA with 30 components fitted on up to 20,000 cells drawn from the full SciPlex3 matrix at seed 0. Each drug's latent delta is the mean over lines of (mean latent of treated − that line's control latent mean), again requiring at least two treated cells. Prediction encodes each real control cell of the query context, adds the drug's latent delta to that cell's own latent code, and decodes each cell by the PCA inverse transform. The decoder is linear, so the composite map is $\text{dec}(\text{enc}(x) + \text{d}z) = \text{P}x + \text{P}\,\text{d}z$ and the predictor is additive in gene space and deterministic. Its zero-effect baseline is the PCA reconstruction of the control cells
scGen	scGen fitted using scvi-tools; trained on the SciPlex3 expression matrix, perturbation as batch and cell line as label; 30 latent dimensions; batch size 512; at most 40 epochs with early stopping after 15 epochs without improvement; seed 0; per-perturbation latent delta added to each encoded control cell and decoded
Optimal transport	Entropic (Sinkhorn) coupling in a 30-component PCA latent; cost rescaled by its maximum; uniform marginals; regularisation 0.05; at most 500 iterations; fitted per non-query context centred on that context's own control mean, with the query context excluded; barycentric projection extended to the query's control cells by nearest-source matching and decoded by the PCA inverse transform

The average-effect, nearest-neighbour and linear-latent predict-then-rank diagnostics use the full SciPlex3 matrix and are interpreted as in-sample candidate-generation references. The entropic optimal-transport diagnostic instead excludes the query context during fitting and is assessed separately for cross-context transfer.

The predict-then-rank experiment used 720 cross-line queries, formed from the three cell-line pairs (K562–A549, A549–MCF7, K562–MCF7) at mixing weights $\alpha \in \{0.5, 0.7, 0.9\}$ over 10 divergent drugs and 8 seeds; 200 query cells per query; a 31-drug candidate library holding the true drug and 30 other shared drugs; 200 predicted cells synthesised per candidate; query construction seeded at $1000 + s$ and distractor selection at $7000 + s$ for $s = 0, \ldots, 7$. The within-context differential-response measurements in Supplementary Note 4 are a separate run over 30 cell-line and drug combinations with 400 control cells per context split into two states, a state requiring 60 cells and a treated arm 40, at seed 0. Nearest-neighbour appears only in the predict-then-rank run and scGen only in the differential-response run; the entropic optimal-transport map appears in both.

Supplementary Table 5. Matched-geometry protein evaluator comparison.

RNA retrieval score	Protein evaluator		
	Direction-only cosine	Magnitude-aware mean L2	Population energy
All queries ($n = 659$)			
Direction-only cosine	+0.2527	+0.2967	+0.2299
Magnitude-aware mean L2	+0.1559	+0.4255	+0.3199
Population energy	+0.2235	+0.4150	+0.3930
Co-culture ($n = 219$)			
Direction-only cosine	+0.3132	+0.3087	+0.2760
Magnitude-aware mean L2	+0.2668	+0.4917	+0.3973
Population energy	+0.2666	+0.4771	+0.4412
Control ($n = 213$)			
Direction-only cosine	+0.2492	+0.2008	+0.1202
Magnitude-aware mean L2	+0.0657	+0.3177	+0.1544
Population energy	+0.2435	+0.3366	+0.3316
IFNγ ($n = 227$)			
Direction-only cosine	+0.2091	+0.3473	+0.2685
Magnitude-aware mean L2	+0.1521	+0.4426	+0.3394
Population energy	+0.1654	+0.4230	+0.3929

Entries are median per-query Spearman correlations across candidate knockouts. Rows are fixed RNA-derived retrieval scores and columns are evaluators constructed from the same held-out protein measurements. The primary matched-geometry comparison is magnitude-aware mean-L2 versus population-energy retrieval under the protein mean-L2 and population-energy evaluators. The median within-query difference (RNA energy minus RNA mean-L2) is -0.0208 under the protein mean-L2 evaluator and $+0.0538$ under the protein population-energy evaluator. It is -0.0105 and $+0.0466$ in Co-culture, -0.0363 and $+0.1342$ in Control, and -0.0318 and $+0.0384$ in IFNγ, respectively. Paired differences are medians of within-query differences, not differences between marginal medians.

Supplementary Table 6. Predicted-response gains against two mean references.

Predictor	Energy minus mean cosine	Energy minus mean L2
Average effect	−0.0356 (−0.0620, −0.0115)	−0.0020 (−0.0036, −0.0005)
Linear latent	−0.0341 (−0.0601, −0.0100)	−0.0016 (−0.0033, +0.0001)
Nearest context	−0.3159 (−0.3544, −0.2754)	−0.0032 (−0.0045, −0.0020)
Nearest context, cells	−0.3399 (−0.3736, −0.3027)	−0.0321 (−0.0501, −0.0151)
Gaussian optimal transport	+0.0027 (−0.0215, +0.0243)	+0.0110 (+0.0087, +0.0134)
State-conditioned average	−0.0314 (−0.0584, −0.0076)	−0.0019 (−0.0037, −0.0002)

Entries are paired MRR differences with 95% percentile intervals from 2,000 cell-line-cluster bootstrap resamples, using 3,992 queries in 44 held-out cell lines, each evaluated at five seeds. Energy uses the U-statistic. Query, candidate population, split and prediction are identical across each paired comparison. Against observed candidates, energy gives +0.0303 (+0.0225, +0.0398) over mean cosine and +0.0102 (+0.0065, +0.0145) over mean L2.

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